PIONEER JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2013

Candidate Name	

Civics Group						**INDEX NUMBER**					

H2 BIOLOGY **9648 / 03**

20 Sept 2013 **2 hours**

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Read the instructions on the answer sheet very carefully.
This document consists of 2 sections:

Section A
Answer **ALL** questions.
Answers are to be written in the spaces provided.

Section B
Answer Question 5.
Write your answer on the lines provided.

Do not open this booklet until you are told to do so.

For Examiner's Use	
Section A	
1	**/ 12**
2	**/ 14**
3	**/ 14**
Section B	
5	**/ 20**
TOTAL	**/ 60**
4 (SPA)	**/12**

This document consists of **22** printed pages (inclusive of this page)

[Turn Over

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Section A: Answer all questions [40 marks]

1. The ability to identify salmon species following food processing such as canning and smoking is problematic but of great commercial importance. The most commonly used methods of identifying raw fish are not applicable to thermally treated products. Polymerase chain reaction-restriction fragment length polymorphism (PCR-RFLP) is thus an alternative method to authenticate canned or smoked fish products.

   (a) Suggest why authentication of processed fish product is important. [1]

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   Following extraction of DNA from the processed fish product, PCR was carried out to amplify a 500bp segment from the DNA sample. Figure 1.1 illustrates the thermal cycling characteristic of PCR.

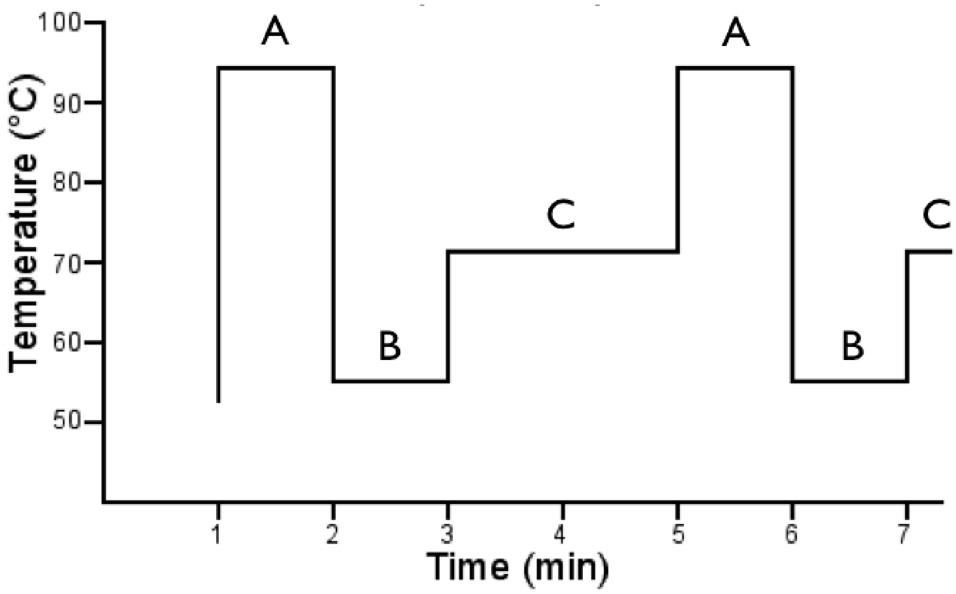

Figure 1.1

   (b) Explain the importance of the temperatures in phases A, B and C.

   (i) Phase A [2]

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(ii) Phase B [1]

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(iii) Phase C [1]

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Figure 1.2 shows the number of molecules of DNA produced over a number of PCR cycles.

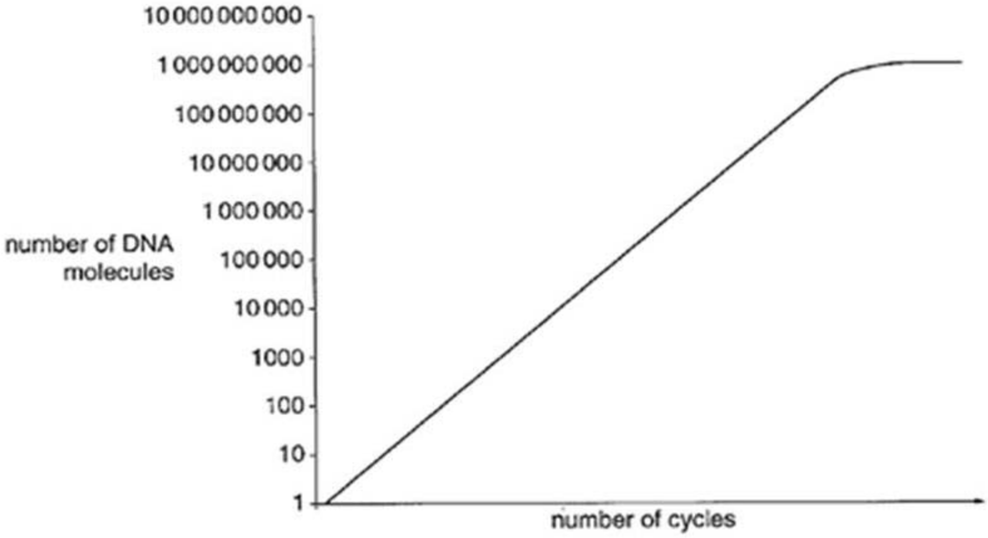

Figure 1.2

(c) Suggest a reason why the graph plateaus. [1]

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DNA samples were taken from 10 different smoked salmon products to authenticate the salmon used. The amplified samples were then subjected to restriction digests using various restriction enzymes. For each reaction, the DNA restriction fragments were subjected to gel electrophoresis and visualized using a staining kit.

Figure 1.3 shows the RFLP profiles of salmon products 1 to 10 following digestion with restriction enzymes Bsp 1286I, Eco RII and Sau 3AI. 100bp DNA ladder lanes were included as reference.

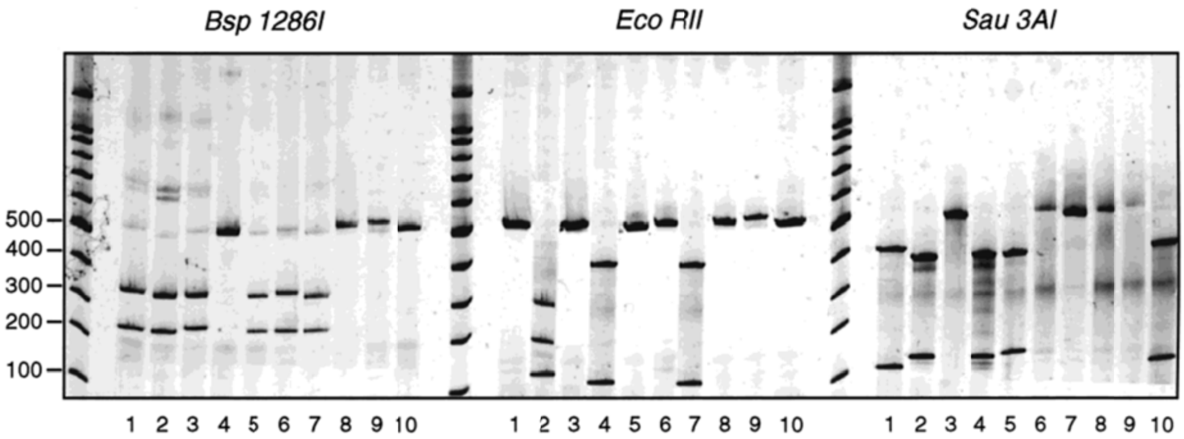

Figure 1.3

With reference to Figure 1.3,

(d) explain what is meant by RFLP. [2]

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(e) identify and explain which 2 smoked salmon products used the same species of salmon. [4]

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[Total: 12 marks]

2. The first gene therapy trial for an inherited disorder was initiated on 14 September 1990. The patient, Ashanthi DeSilva, suffered from a very rare recessively inherited disorder, adenosine deaminase (ADA) deficiency.

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An inherited deficiency of ADA has, however, particularly severe consequences in the case of T lymphocytes. As a result, ADA-deficient patients suffer from severe combined immunodeficiency (SCID). #

(a) Explain the importance of ADA with respect the immune system. [2]

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(b) Suggest why a recessively inherited disorder such as ADA-SCID can be treated by gene therapy. [2]

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(c) The novel ADA gene therapy approach involved essentially the following steps:

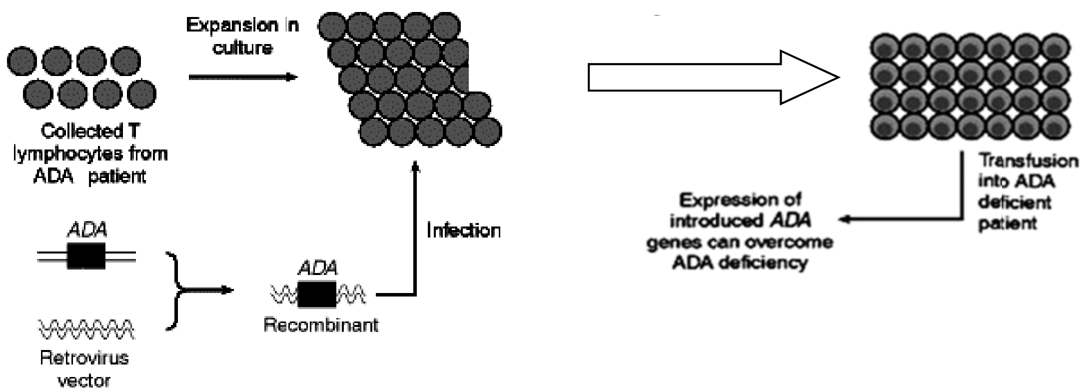

Figure 2.1

(i) Suggest why the *ex vivo* method was used in this case. [1]

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(ii) Suggest an improvement to the procedure in Figure 2.1 so as to increase the success rate of the treatment. [3]

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(d) Name another method, other than gene therapy, which can be used for the treatment of ADA-deficiency. [1]

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(e) Gene therapy using the same form of treatment as the above case study has succeeded for 15 French baby boys suffering from X-linked SCID. However 3 of the little boys later developed cancer and died. Suggest how this is possible. [3]

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(f) State and explain 2 other factors that hinder the effectiveness of current gene therapy approaches. [2]

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[Total: 14 marks]

3. AquAdvantage salmon is a transgenic salmon where a promoter for the anti-freeze gene of an ocean pout and a gene for the Chinook growth hormone was inserted into a plasmid pUC18 (Figure 3.1). The recombinant DNA (Figure 3.2) is then introduced into fertilized egg cells of the Atlantic salmon by microinjection and stimulated to become triploid.

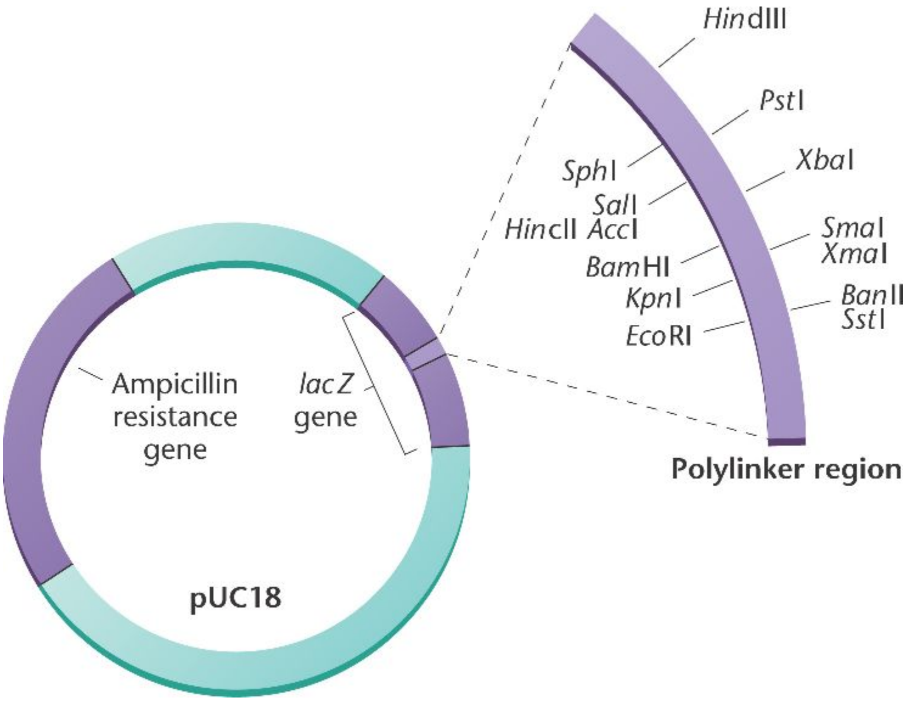

Figure 3.1

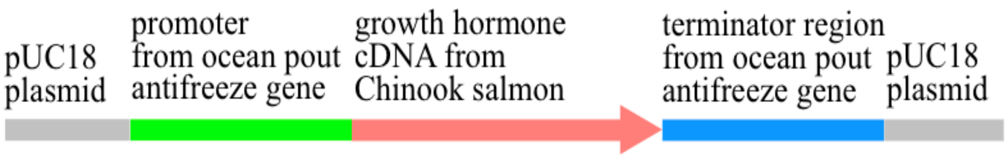

Figure 3.2

(a) With reference to the Figure 3.1 and 3.2,

(i) describe the role of the ampicillin resistance gene. [1]

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(ii) describe how the recombinant DNA is formed. [4]

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(b) The egg cells were induced to be triploid. Explain the significance of this. [2]

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(c) Figure 3.3 below shows the differences between weight of AquAdvantage salmon and standard salmon.

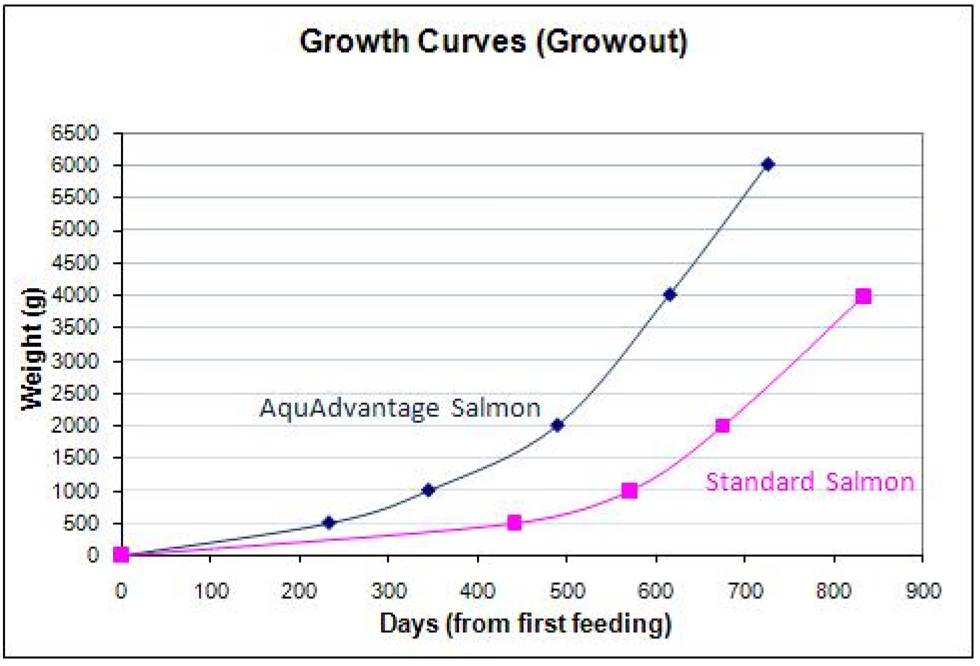

Figure 3.3

(i) With reference to Figure 3.3, compare the growth curves of standard salmon and AquAdvantage salmon. [3]

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(ii) Explain the difference in the weight increase of AquAdvantage salmon and the standard salmon. [2]

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(d) Describe one ethical and one social concern that might arise from the production of AquAdvantage salmon. [2]

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[Total: 14 marks]

4. **Planning Question [12 marks]**

The Tasmanian Tiger or Thylacine (*Thylacinus cynocephalus*), the world's largest carnivorous marsupial, was once common throughout Australia and Papua New Guinea. The Thylacine resembled a large, short-haired dog with a stiff tail which smoothly extended from the body in a way similar to that of a kangaroo. 7ªe female Thylacine had a pouch with four teats, but unlike many other marsupials, the pouch opened to the rear of its body.#

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An example of convergent evolution, the Thylacine showed many similarities to the members of the Canidae (dog) family of the Northern Hemisphere: sharp teeth, powerful jaws, raised heels and the same general body form.#

Due to human activities, the Thylacine was hunted to extinction by early 1930. A Thylacine specimen with soft tissue remaining is found in the Australian Museum in Sydney.

Imagine that you are a researcher in the Australian Rare Fauna Research Association. Recently it was reported that locals near Mount Carstensz in Western New Guinea had sighted creatures that resemble Thylacines. Some members of the Association believe that the creatures sighted may be descendents of the Thylacine, while other members believe that the creatures may be a new species. If the former were true, then the Thylacine is not extinct and conservation effects may revive the species.

Plan an investigation to investigate if these creatures found in Western New Guinea are descendents of the Thylacine that was thought to be extinct.

Your planning must be based on the assumption that you have been provided with the following equipment and materials.

- Tissue sample from the museum specimen and from the Western New Guinea creatures under investigation
- Pestle and mortar
- DNA extraction buffer solution
- Microcentrifuge tubes
- Centrifuge
- Restriction enzymes
- Agarose or polyacrylamide gel plate
- Suitable source of electric current
- Radioactive probe
- Nitrocellullose membrane
- Autoradiography equipment

Your plan should have a clear and helpful structure to include:

- An explanation of the theory to support your practical procedure
- A description of the method used including scientific reasoning behind the method
- The type of data generated by the experiment
- How the results will be analysed including how the origin of the organism can be determined

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Section B: Essay Question [20 marks]

Write your answers to this question on the lines provided.

Your answer:

- Should be illustrated by large, clearly labelled diagrams, where appropriate.
- Must be in continuous prose, where appropriate.
- Must be set out in section (a), (b), etc., as indicated in the question.

(a) Describe how the micropropagation technique is used in production and cloning of herbicide resistant plants. [9]
(b) Discuss the social and ethical implications of GMO plants using 3 named examples. [6]
(c) Discuss the limitations for the use of gene therapy to treat diseases in humans. [5]
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